4 November 2026

Professor Jing-Zhi Huang
Editor-in-Chief
The Journal of Finance and Data Science

Subject: Submission of a research article for consideration

Dear Professor Huang,

We would like to submit our manuscript, "*Beyond Directional Accuracy: A Verified Walk-Forward Framework for Machine-Learning Cryptocurrency Trading*", for consideration as a research article in *The Journal of Finance and Data Science*.

Many studies that use machine learning to predict cryptocurrency prices report only directional accuracy. In our work we saw that accuracy alone can give a wrong picture of a model. So we built a step-by-step evaluation framework that connects forecasting, statistical testing and trading costs, and that also checks the backtest code itself with tests. We used it on daily Binance data for Bitcoin, Ether and Dogecoin, with 5,562 out-of-sample forecasts from July 2020 to March 2026.

The main findings are:

- A regularized logistic regression model is correct 52.95% of the time, and it is better than a coin flip on all three assets.
- A simple one-day reversal rule gets almost the same accuracy, but the model is better on the large price moves. Its long/short portfolio earns a gross Sharpe ratio of 0.54, compared with 0.004 for the reversal rule.
- With a cost of 9 basis points per unit of turnover, the net Sharpe ratio stays positive (0.11), and the break-even cost is 11.4 basis points.
- The tests of the backtest engine caught all eight rule violations that we put back into the code on purpose.

We think the paper fits the aims and scope of the journal well. It uses data science methods for a finance problem, and it gives a practical method that other researchers can use and check. All forecasts are saved with their dates, and the full analysis can be run again with one command. The journal has published related work on bitcoin prediction with machine learning (Jaquart et al., 2021), and our paper adds to this line of work with an evaluation that takes simple baselines, the size of price moves and trading costs into account.

We confirm that this manuscript is our own original work. It has not been published before, and it is not under consideration at any other journal. Both authors have read and approved the final version and agree with this submission. The authors have no competing interests, and this research did not receive any specific funding. The study uses only public market data, so ethics approval was not needed. We used AI tools in some parts of the work, and we explain this in the declaration of generative AI at the end of the manuscript.

Because the journal uses double-anonymized peer review, the manuscript file does not show the names of the authors. The author details, the declarations and the link to the code and data are given on the separate title page.

Thank you for your time and for considering our manuscript. We look forward to hearing from you.

Yours sincerely,

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